

Doyun Kim

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SUMMARY

- Experienced AI and ML Engineer with a strong background in circuit design.
- Proven leadership in managing and executing interdisciplinary research projects.
- Keen interest in applying AI for design productivity and optimization in semiconductor industry.

EXPERIENCE

Senior AI Engineer, Normal Computing

May 2025 ~ Present

- Spec-to-SystemVerilog Assertion Generation Solution
 - Architected and deployed LLM-based workflow utilizing function calling for structured intermediate representation, enabling automated assertion generation from specifications
 - Built AI agent for testbench analysis and SystemVerilog assertion module implementation
 - Successfully delivered proof-of-concept to client, demonstrating feasibility of AI-driven verification automation
- Benchmark Framework for Design Verification
 - Established evaluation infrastructure for AI agents using OpenTitan open-source digital design data
 - Developed containerized benchmark server enabling reproducible assessment of design verification tasks

Project Leader, Design AI Lab, Samsung Electronics

Jan 2025 ~ Apr 2025

- AI Strategy Development
 - Developed AI initiatives with senior executives to nurture long-term AI competitiveness and align with overall business objectives.
 - Established an AI committee with business unit leaders and co-developed a strategic roadmap to drive AI innovation in semiconductor design.
- Project Management
 - Led multiple technical projects including DRAM Peripheral Circuit Placement, DRAM Bitline Sense Amplifier Optimization, DRAM Standard Cell Optimization, SRAM Peripheral Circuit Placement, and SSD Test Scenario Generation.
 - Applied AI technology in DRAM, SSD and Foundry product design process to enhance product quality and reduce time to market.
- People & Resource Management
 - Oversaw budget allocation and execution
 - facilitating partnerships with academic institutions to support cutting-edge research.
 - Directed recruitment efforts to build and nurture a high-performing, multidisciplinary technical team.

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Activity Leader, Samsung Advanced Institute of Technology

Mar 2024 ~ Dec 2024

- Graph Foundation Model for Semiconductor Design
 - Led to develop Graph-Transformer model and a novel Positional Encoding to represent circuits
 - Improved representation power, resulting in $R^2=0.88$ for predicting analog circuit performance
- Bitline Sense Amplifier (BLSA) optimization under process variation
 - Optimized circuit and timing parameters of DRAM core using Reinforcement Learning for robust design against mismatch.
 - Achieved ~10% robustness improvement and 56% Turn-around time reduction

Principal Research Scientist, Samsung Advanced Institute of Technology

Feb 2022 ~ Feb 2024

- SSD PCB Trace S-parameter Prediction
 - Developed a PCB data generator with product-level design rules and constraints
 - Built S-parameters (2401 x 2 x 210 dimension) prediction model based on Graph Transformer, utilizing geometrical and topological features from graph representations of PCB traces, achieving $R^2>0.85$ up to 15GHz
- SSD DRAM I/O Performance and Dominant Factor Prediction based on Tabular Data
 - Performed analysis and pre-processing on SSD firmware data sets
 - Built a prediction system with Tab-Transformer and XGBoost models to monitor DRAM I/O performance in real-time, achieving ($R^2>0.65$)
 - Identified dominant factors for the I/O performance result based on SHAP analysis
- Analog Circuit Optimization
 - Supported development of model-based reinforcement learning algorithms for analog circuits (OpAmp, TIA, Comparators) by setting up opensource-based circuit simulation environments
 - Proposed a neural simulator to estimate performance distributions under local variation and reward design for robust design without numerous MC simulations
 - Achieved 5x sample efficiency to meet the target specifications under local variation

Research Scientist, Facebook

Mar 2019 ~ Dec 2021

- Power delivery system optimization framework in Python
 - Optimized PMIC architecture (power tree) using machine learning clustering techniques
 - Built python models for PMIC components including regulators and power rails
 - Implemented web dashboard to visualize power across different PMIC configurations and use cases
- Design custom SRAM for hardware accelerator application
 - Designed column drivers and wordline drivers with write-assist technique in 7nm

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- Analyzed the impacts of design parameters on SRAM power, latency and error rate
- Implemented scripts language to run multiple SPICE simulation with different configuration bits

Graduate Research Assistant, Columbia University

Sep 2013 ~ Dec 2018

- SRAM-ADC
 - Designed a novel SRAM architecture which can transform into a stochastic ADC with 6-bit level in 65nm
 - Repurposed bitcells as gain stages and linearized ADC output by pre-characterization with <1% area overhead
- Event-driven LDO designs based on digital PI controller
 - Designed continuous-time ADC and trigger generator for latency reduction
 - Implemented and optimized the digital controller algorithm to accelerate voltage regulation
 - Achieved 74x performance improvement in FoM and 4x area reduction over 3 prototyped designs in 65nm
- Analytical Model of Energy-Optimal Voltage
 - Derived an analytical solution of energy-optimal voltage by key technology parameters for V_{DD} scaling
 - Modeled the critical activity factor at which the energy-optimal voltage starts to saturate
 - Verified the models with <10% error across different technology nodes (130nm-45nm) in simulation

EDUCATION

Oct 2019	Columbia University (Advisor: Mingoo Seok) <i>M.S./Ph.D. in Electrical Engineering, GPA 3.87</i> Thesis: Fully Integrated Digital Low-drop-out Regulator Design based on Event-Driven PI control
Feb 2013	Pohang University of Science and Technology (POSTECH) <i>B.S. in Electronic and Electrical Engineering, GPA 3.90 (summa cum laude)</i> (Exchange student program in University of Minnesota, Twin cities, Fall 2011)

SKILLS

- **Circuit Design:** Proficiency in Verilog, Virtuoso, Spectre, HSPICE, SIWave
- **AI development:** Experience in PyTorch, Custom AI Agent
- **Language:** Python, Bash, Tcl

- 16 T Paik, J Park, T Kim, DH Jung, **D Kim**, "PCBFormer: Understanding 3D Structure of RealWorld PCB Traces for S-Parameter Prediction", 2025 IEEE/ACM International Conference On Computer Aided Design (ICCAD), 2025
- 15 S Kim, Z Wang, S Lee, Y Oh, H Zhu, **D Kim**, DZ Pan, "Ppaas: Pvt and pareto aware analog sizing via goal-conditioned reinforcement learning" IEEE/ACM International Conference On Computer Aided Design (ICCAD), 2025
- 14 MG Park, **D Kim**, SK Lim, "MIX-3D: AI-based Architecture-Circuit Co-design Methodology for Mixed-Node, Mixed-Area 3D ICs" ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED), 2025
- 13 YH Lee, J Park, T Paik, **D Kim**, B Hwang, "From Theory to Practice: Rethinking Green and Martin Kernels for Unleashing Graph Transformers" International Conference on Machine Learning (ICML), 2025
- 12 **D. Kim**, J. Park, Y. Oh, and B. Hwang, "TraceFormer: S-parameter Prediction Framework for PCB Traces based on Graph Transformer," *ACM/IEEE Design Automation Conference (DAC)*, 2024
- 11 Y. Oh, **D. Kim**, Y. Lee, and B. Hwang, "CRONuS: Circuit Rapid Optimization with Neural Simulator," *IEEE Design, Automation & Test in Europe Conference & Exhibition (DATE)*, 2024
- 10 **D. Kim**, P. R. Kinget, and M. Seok, "SRAM-ADC: SRAM Circuits Transformable to a Stochastic ADC at Ultralow Area Overhead," *IEEE Solid-State Circuits Letters (SSCL)*, 2019
- 9 T. Yang, **D. Kim**, J. Li, P. R. Kinget and M. Seok, "In Situ and In-Field Technique for Monitoring and Decelerating NBTI in 6T-SRAM Register Files," *IEEE Transactions on Very Large Scale Integration Systems (TVLSI)*, 2018
- 8 M. Seok, P. R. Kinget, T. Yang, J. Li and **D. Kim**, "Recent advances in in-situ and in-field aging monitoring and compensation for integrated circuits: Invited paper," *IEEE International Reliability Physics Symposium (IRPS)*, 2018
- 7 **D. Kim**, S. Kim, H. Ham, J. Kim and M. Seok, "0.5V- V_{IN} , 165-mA/mm² Fully-Integrated Digital LDO based on Event-Driven Self-Triggering Control," *Symposium on VLSI Circuits (VLSIC)*, 2018
- 6 **D. Kim** and M. Seok, "A Fully Integrated Digital Low-Dropout Regulator Based on Event-Driven Explicit Time-Coding Architecture," *IEEE Journal of Solid-State Circuits (JSSC)*, Nov. 2017
- 5 S. J. Kim, **D. Kim** and Mingoo Seok, "Comparative study and optimization of synchronous and asynchronous comparators at near-threshold voltages," *IEEE/ACM International Symposium on Low Power Electronics and Design (ISLPED)*, 2017

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- 4 **D. Kim**, J. Kim, H. Ham, and M. Seok, "A 0.5V- V_{IN} 1.44mA-Class Event-Driven Digital LDO with a Fully Integrated 100pF Output Capacitor," *IEEE International Solid-State Circuits Conference (ISSCC)*, 2017
- 3 **D. Kim** and M. Seok, "Fully-Integrated Low Drop-Out Regulator based on Event-Driven PI Control," *IEEE International Solid-State Circuits Conference (ISSCC)*, 2016
- 2 **D. Kim**, J. Li, and M. Seok, "Energy-Optimal Voltage Model Supporting a Wide Range of Nodal Switching Rates for Early Design-Space Exploration" *IEEE International Conference on Computer Design (ICCD)*, 2015
- 1 T. Yang, **D. Kim**, P. R. Kinget, and M. Seok, "In-situ techniques for in-field sensing of NBTI degradation in an SRAM register file," *IEEE International Solid-State Circuits Conference (ISSCC)*, 2015

Professional Services

- Conference Organization and Technical Program Committee
 - Technical Program Committee Member, Design Automation Conference, 2024-2026
- Editorial Work
 - Journal Reviewer, IEEE Journal of Solid-State Circuits, 2019-2021
 - Journal Reviewer, IEEE Transactions on Very Large Scale Integration (VLSI) Systems, 2015
 - Journal Reviewer, IEEE Transactions on Circuits and Systems I: Regular Papers, 2014